

Volume 9 – UI/UX Specification & Design System

AI Quant Trading Platform

Version: 1.0

Document Type: UI/UX Specification

Classification: Frontend Architecture

1. Purpose

This document defines the complete user interface architecture, user experience guidelines, navigation flow, component library, responsive layouts, accessibility requirements, and screen specifications for the AI Quant Trading Platform.

The goal is to build a professional trading terminal that is intuitive for beginners while providing advanced capabilities for experienced traders.

2. Design Philosophy

The interface shall prioritize:

- Clarity
- Speed
- Minimal clicks
- Real-time responsiveness
- Explainable AI
- Data visualization
- Consistency
- Accessibility

Every screen should help users make informed decisions rather than overwhelm them with information.

3. Design System

Typography

- Primary Font
- Secondary Font
- Monospace Font for trading values

Color Roles

- Primary
- Secondary
- Success
- Warning
- Error
- Neutral

Theme Support

- Light Theme
- Dark Theme
- High Contrast Theme (future)

Spacing System

- 4px Grid
 - Consistent padding and margins
 - Responsive containers
-

4. Navigation Structure

Main Navigation

Dashboard



Market Scanner



Trading Terminal



Portfolio



Strategies



AI Insights



Reports



Notifications



Settings



Support

Admin Navigation

Users



Subscriptions



System Health



Workers



Logs



Configuration

5. User Roles

Guest

Registered User

Premium User

Enterprise User

Support Staff

Admin

Super Admin

Each role sees only the features and navigation relevant to its permissions.

6. Screen Inventory

Authentication

- Welcome
 - Login
 - Register
 - Forgot Password
 - Reset Password
 - Verify Email
 - Two-Factor Authentication
 - Session Management
-

Dashboard

Widgets

- Portfolio Value
 - Daily P&L
 - Weekly P&L
 - Monthly P&L
 - Active Positions
 - Market Overview
 - AI Confidence Summary
 - Top Opportunities
 - Open Orders
 - Strategy Status
 - Risk Exposure
 - Notifications
-

Market Scanner

Features

- Live Coin List

- Search
 - Filters
 - AI Score
 - Confidence Score
 - Trend
 - Volume
 - Market Regime
 - Watchlist
 - Compare Assets
-

Trading Terminal

Components

- Trading Chart
 - Order Entry
 - Position Panel
 - Order Book
 - Recent Trades
 - AI Recommendation
 - Technical Indicators
 - Risk Estimate
 - Strategy Details
-

Portfolio

Views

- Holdings
 - Asset Allocation
 - Performance
 - Realized P&L
 - Unrealized P&L
 - Trade History
 - Exposure
 - Diversification
-

Strategy Center

Features

- Strategy Library
- Enabled Strategies
- Configuration
- Parameters
- Historical Performance

- Backtesting
 - Paper Trading
-

AI Insights

Features

- Opportunity Ranking
 - Confidence Breakdown
 - Supporting Factors
 - Market Regime
 - Trend Analysis
 - AI Reasoning
 - Historical Decisions
-

Reports

Reports

- Daily
 - Weekly
 - Monthly
 - Yearly
 - Tax Export
 - Strategy Performance
 - Trade Journal
-

Notifications

Categories

- Trading
 - Portfolio
 - Risk
 - Exchange
 - Billing
 - System
-

Settings

Sections

- Profile
- Security

- Exchanges
 - Strategies
 - Notifications
 - Appearance
 - API Keys
 - Billing
-

Admin Portal

Screens

- User Management
 - Subscription Management
 - Exchange Monitoring
 - Queue Status
 - Worker Status
 - Audit Logs
 - Feature Flags
 - System Configuration
 - Analytics
-

7. Dashboard Layout

Header

- Logo
- Search
- Notifications
- User Menu

Sidebar

- Navigation
- Quick Actions

Main Content

- Widgets
- Charts
- Tables
- AI Panels

Right Panel

- Alerts
- Activity Feed
- Market News

Footer

- System Status
 - API Status
 - Version
-

8. Widget Library

Widgets

- Market Ticker
 - AI Score Card
 - Position Card
 - P&L Card
 - Portfolio Pie Chart
 - Performance Graph
 - Strategy Card
 - Risk Meter
 - Notification Feed
 - Economic Calendar
 - Exchange Status
-

9. Trading Chart

Capabilities

- Multiple Timeframes
 - Drawing Tools
 - Indicators
 - AI Overlay
 - Entry Zones
 - Exit Zones
 - Support & Resistance
 - Trade History
 - Risk Visualization
-

10. Forms

Every form shall support:

- Inline Validation
- Keyboard Navigation
- Autosave where appropriate
- Error Recovery

- Loading Indicators
 - Confirmation Dialogs
-

11. Tables

Features

- Pagination
 - Sorting
 - Filtering
 - Export
 - Column Selection
 - Bulk Actions
 - Sticky Headers
-

12. Search

Global Search

Search:

- Coins
- Strategies
- Reports
- Trades
- Users (Admin)
- Exchanges

Support keyboard shortcuts.

13. Responsive Design

Desktop

Primary platform.

Tablet

Optimized layout.

Mobile

Essential features only.

Responsive breakpoints must maintain usability without hiding critical trading information.

14. Accessibility

Requirements

- WCAG compliance where practical
 - Keyboard navigation
 - Screen reader support
 - Focus indicators
 - Color-independent status indicators
 - Adjustable font scaling
-

15. User Experience Guidelines

- Minimize clicks for common tasks.
 - Highlight important risk information.
 - Use progressive disclosure for advanced settings.
 - Display loading states for asynchronous actions.
 - Confirm irreversible actions.
 - Provide contextual help.
-

16. Component Library

Core Components

- Buttons
- Inputs
- Dropdowns
- Modals
- Tabs
- Accordions
- Cards
- Tables
- Charts
- Toasts
- Badges
- Avatars
- Breadcrumbs
- Tooltips
- Skeleton Loaders

Reusable components must be shared across all modules.

17. Design Tokens

Define reusable tokens for:

- Colors
 - Typography
 - Border Radius
 - Shadows
 - Animation Durations
 - Spacing
 - Icon Sizes
-

18. Animations

Use subtle animations for:

- Data refresh
- Notifications
- Loading states
- Drawer transitions
- Modal appearance

Animations must not delay critical trading interactions.

19. Internationalization

Support:

- Multiple Languages
 - Time Zones
 - Date Formats
 - Number Formats
 - Currency Formats
-

20. Error Handling

User-friendly error states:

- Exchange Offline
- Network Failure
- Authentication Expired
- Validation Errors
- Rate Limit Reached
- Market Data Unavailable

Provide recovery guidance where possible.

21. Empty States

Examples:

- No Trades Yet
- No Connected Exchange
- No Notifications
- No Strategy Enabled

Include clear actions to help users proceed.

22. Performance Targets

- Initial dashboard load under 2 seconds (typical conditions)
 - Real-time updates without full-page refresh
 - Efficient rendering for large watchlists
 - Lazy loading for heavy components
-

23. Future Enhancements

- Custom dashboard layouts
 - Drag-and-drop widgets
 - Multi-monitor mode
 - Command palette
 - Voice assistant
 - AI chat panel
 - Collaborative workspaces
 - Plugin marketplace
-

24. Design Principles

- Every important action should be reachable within a few interactions.
- Real-time information should remain readable and stable.
- AI recommendations must always include context and explanation.
- Risk information should be visually prominent.
- Interfaces should scale from beginner to professional users without requiring separate applications.